

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

CO1-AT1-Analytical Problem Solving

Register Number	192511025
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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 40%

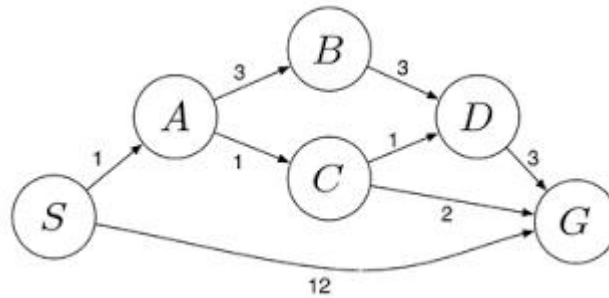
QUESTIONS: 5

Total Marks: 50

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost

path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



Code of Conduct:

I S.Swedha/192511025 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Swedha

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Problem Understanding	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	10
Method Selection	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	10
Solution Process	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	12.5
Accuracy of Calculation	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	10
Interpretation of Results	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	7.5

Answers:

1. The Water Jug Problem is one of the classic examples used in Artificial Intelligence (AI) to demonstrate problem-solving using state-space search. It is commonly solved using Uninformed Search techniques such as Breadth-First Search (BFS) or Depth-First Search (DFS). The problem involves finding a sequence of valid actions that transforms the initial state into the desired goal state. In this problem, the agent has no prior knowledge about the solution and explores different possible states until it reaches the goal.

Problem formulation is the process of defining the components required for solving an AI problem.

Initial State:

Both jugs are empty.

$(0,0)$

Where ,

First value represents water in the 4-gallon jug

Second value represents water in the 3-gallon jug

Goal State

The objective is to obtain exactly 2 gallons of water in the 4-gallon jug.

Goal State:

$(2, x)$

where x may be 0 or any permissible amount.

In this solution, the goal state is

$(2,0)$

Each state represents :

(4 gallon jug , 3 gallon jug)

The following operations can be performed at any state:

1. Fill the 4-gallon jug completely.
2. Fill the 3-gallon jug completely.
3. Empty the 4-gallon jug.
4. Empty the 3-gallon jug.
5. Pour water from the 4-gallon jug into the 3-gallon jug.
6. Pour water from the 3-gallon jug into the 4-gallon jug.

These actions generate new states during the search process

Step	Action Performed	State (4G,3G)	Explanation
Initial	Both jugs are empty	(0,0)	Starting state
1	Fill the 3-gallon jug	(0,3)	Pump fills the smaller jug completely
2	Pour water into the 4-gallon jug	(3,0)	All 3 gallons are transferred
3	Fill the 3-gallon jug again	(3,3)	Smaller jug is filled again
4	Pour water into the 4-gallon jug until it becomes full	(4,2)	Only 1 gallon is needed to fill the larger jug, so 2 gallons remain in the smaller jug
5	Empty the 4-gallon jug	(0,2)	The larger jug is emptied
6	Transfer the remaining 2 gallons into the 4-gallon jug	(2,0)	Goal state is achieved

Using the Breadth-First Search strategy, the sequence of actions successfully produces exactly 2 gallons of water in the 4-gallon jug, satisfying the problem requirements.

2. A Mars Rover is an autonomous robotic vehicle developed to explore the surface of Mars. Since direct human control is not possible due to the long communication delay between Earth and Mars, the rover must act as an intelligent agent. It continuously senses its environment, processes the collected information, makes decisions, and performs suitable actions to accomplish its mission.

The main objectives of a Mars Rover include navigating safely, collecting rock and soil samples, analyzing environmental conditions, and transmitting scientific data back to Earth.

According to Artificial Intelligence, an intelligent agent perceives its environment through sensors and performs actions through actuators to achieve its goals efficiently.

- i) A **percept** is any information that an intelligent agent receives from its environment through its sensors. The Mars Rover continuously gathers different types of data to understand its surroundings.

The percepts of the Mars Rover include:

- Images captured by high-resolution cameras.
- Distance from nearby rocks and obstacles using obstacle detection sensors.
- Surface temperature measurements.
- Atmospheric pressure and weather conditions.
- Soil composition and mineral content.
- Rock composition obtained through scientific instruments.
- Position and orientation of the rover.
- Battery charge level.
- Wheel movement and traction information.
- Signals received from Earth.

These percepts help the rover understand the current environmental conditions and make appropriate decisions.

- ii) The operating environment of the Mars Rover can be classified based on the properties of AI environments.

Environment Property	Description
Partially Observable	The rover cannot observe every part of Mars at once. It only receives information from its sensors, so some parts of the environment remain unknown.
Stochastic	The environment is unpredictable. Dust storms, uneven terrain, and weather changes may affect the rover's movement and decisions.
Sequential	Every action performed by the rover influences its future actions. For example, choosing an unsafe path may affect future navigation.
Dynamic	The environment changes over time due to weather conditions, dust accumulation, and changing sunlight levels.
Continuous	The rover moves continuously across the Martian surface, and measurements such as speed, distance, and temperature are continuous values.
Single-Agent	The Mars Rover performs its mission independently without competing with other intelligent agents.

Therefore, the Mars Rover operates in a partially observable, stochastic, sequential, dynamic, continuous, and single-agent environment.

iii) Actions performed by Rover

After analysing the information received from its sensors, the Mars Rover performs various actions to complete its scientific mission.

The important actions include:

- Moving forward to explore new locations.
- Moving backward whenever necessary.
- Turning left or right to avoid obstacles.
- Stopping when an obstacle is detected.
- Capturing high-resolution images of rocks and landscapes.
- Collecting rock samples.
- Drilling into rocks for detailed analysis.
- Collecting soil samples.
- Measuring environmental conditions such as temperature and pressure.
- Performing chemical analysis of collected samples.
- Sending scientific data and images to Earth.
- Recharging its battery using solar panels whenever sunlight is available.

These actions allow the rover to explore Mars safely while collecting valuable scientific information.

iv) The performance measure defines how successfully the intelligent agent completes its assigned tasks.

The performance of the Mars Rover can be evaluated using the following criteria:

a) Successful Navigation

The rover should travel safely without colliding with rocks or getting trapped in sand.

b) Scientific Data Collection

The rover should collect accurate rock, soil, and atmospheric data from different locations.

c) Accuracy of Analysis

The collected samples should be analyzed correctly using onboard scientific instruments.

d) Energy Efficiency

Since the rover depends mainly on solar energy, it should minimize unnecessary movement and conserve battery power.

e) Mission Completion

The rover should successfully complete all assigned exploration tasks within the available operational time.

f) Reliable Communication

Scientific observations and collected data should be transmitted successfully to Earth with minimal data loss.

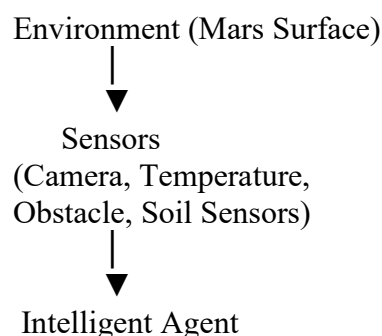
A rover that performs all these tasks efficiently is considered a successful intelligent agent.

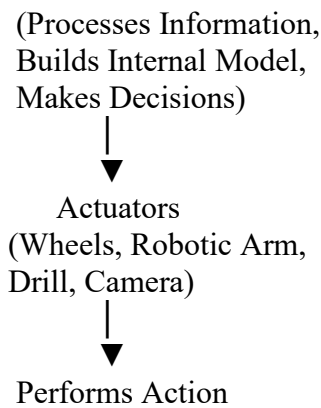
v) A Model-Based Goal-Based Agent combines both memory and planning.

It maintains an internal model of the Martian environment and continuously updates it using information collected from sensors. Based on this model, it selects actions that help achieve mission goals such as reaching a target location, avoiding obstacles, collecting samples, and conserving energy.

For example, if the rover detects a large rock blocking its path, it updates its internal map, plans an alternative route, and safely reaches the destination without human intervention.

Hence, a Model-Based Goal-Based Agent is the most suitable architecture for the Mars Rover





3) The 8-Queens Problem is one of the classical problems in Artificial Intelligence used to demonstrate state-space search, constraint satisfaction, and problem-solving techniques. The objective is to place eight queens on an 8×8 chessboard in such a way that no queen attacks another queen.

In the game of chess, a queen can move horizontally, vertically, and diagonally. Therefore, a valid solution must ensure that no two queens are placed in the same row, same column, or same diagonal.

This problem is commonly solved using Backtracking Search, which is an efficient form of Depth-First Search (DFS).

The initial state is an empty chessboard, where no queens have been placed. A state represents a particular arrangement of queens on the chessboard.

Each time a queen is placed safely, a new state is generated.

The state space consists of all possible arrangements of queens.

Without any restrictions,

Number of possible arrangements

$$= 8^8$$

$$= 16,777,216 \text{ states}$$

However, if only one queen is allowed in each column,

Number of possible arrangements

$$= 8! = 40,320 \text{ states}$$

Backtracking further reduces the search by eliminating invalid states.

The goal state is reached when

- All 8 queens are placed.
- No two queens share the same row.
- No two queens share the same column.
- No two queens share the same diagonal.

Backtracking is an improved version of DFS.

Working:

1. Place a queen.
2. Check if it is safe.
3. If safe, place the next queen.
4. If not safe, remove the queen.
5. Return to the previous column.
6. Try another position.

This avoids exploring unnecessary states.

Therefore, Backtracking is the most efficient search strategy for the 8-Queens Problem.

4) Artificial Intelligence is widely used in modern transportation systems to provide smart and efficient services. One such application is the OLA Cab Booking System, where an intelligent agent assists users in booking a suitable cab based on their requirements.

The agent receives the customer's request, checks the available cab options, compares different alternatives, and selects the most suitable cab to achieve the customer's goal of reaching the destination safely and efficiently.

The OLA booking system is therefore an excellent example of a Goal-Based Intelligent Agent.

The OLA application acts as an intelligent system that continuously interacts with users and drivers. It performs several tasks automatically, including:

- Detecting the customer's location.
- Finding nearby drivers.
- Calculating the estimated fare.
- Estimating arrival time (ETA).
- Suggesting different cab categories.
- Booking the selected cab.
- Tracking the journey until the destination.

The entire booking process is performed automatically using Artificial Intelligence algorithms.

The most suitable intelligent agent for the OLA Cab Booking System is a Goal-Based Agent.

Definition

A Goal-Based Agent is an intelligent agent that selects actions based on a predefined goal. Instead of reacting only to the current situation, it evaluates different possibilities and chooses the action that best helps achieve the desired goal.

The Goal-Based Agent performs the following sequence of operations:

Step 1

The customer opens the OLA application.

Step 2

The agent detects the current location using GPS.

Step 3

The customer enters the destination.

Step 4

The agent searches for nearby drivers.

Step 5

Different cab categories such as Mini, Micro, Sedan, Prime, Auto, and Shared are displayed.

Step 6

The estimated fare and travel time for each cab are calculated.

Step 7

The customer selects the preferred cab.

Step 8

The system assigns the nearest available driver.

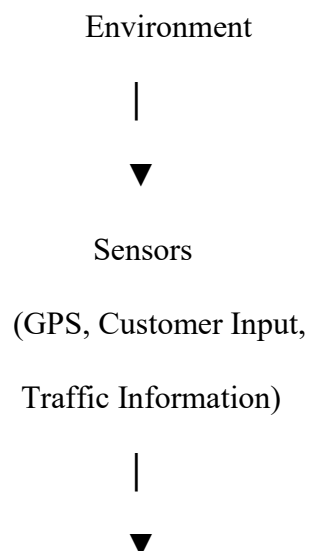
Step 9

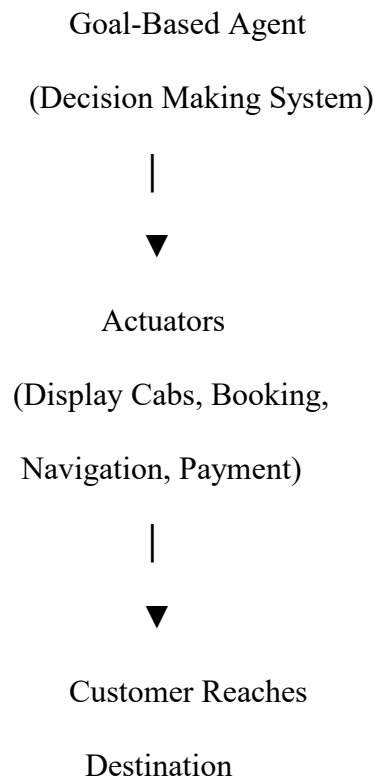
The customer tracks the ride using GPS until reaching the destination.

Step 10

The payment is completed and the trip ends.

Architecture of goal based management :





5) In Artificial Intelligence, many real-world problems require finding the minimum-cost path from a starting point to a destination. Such problems are called path-finding or route optimization problems.

One of the most efficient uninformed search algorithms used to solve these problems is Uniform Cost Search (UCS).

Unlike Breadth-First Search (BFS), which expands nodes level by level, UCS expands the node with the lowest cumulative path cost first. Therefore, it guarantees the optimal solution when all edge costs are positive.

In this problem, the logistics company uses UCS to determine the cheapest route for delivering packages from warehouse S (Start) to warehouse G (Goal).

Uniform Cost Search is an uninformed search algorithm that always selects the node with the smallest accumulated cost for expansion.

Instead of considering the number of steps, UCS considers the total path cost from the starting node.

Working Principle

1. Start from the source node.
2. Insert the source node into the priority queue.
3. Expand the node having the smallest cumulative cost.
4. Add all its child nodes along with their cumulative costs.
5. Continue the process until the goal node is removed from the priority queue.
6. The first time the goal node is expanded, the optimal path is obtained.

State Space

The state space consists of all possible paths connecting the warehouses.

Possible states include:

- S
- $S \rightarrow A$
- $S \rightarrow G$
- $S \rightarrow A \rightarrow B$
- $S \rightarrow A \rightarrow C$
- $S \rightarrow A \rightarrow C \rightarrow D$
- $S \rightarrow A \rightarrow C \rightarrow G$
- $S \rightarrow A \rightarrow B \rightarrow D \rightarrow G$

Step Expanded Node Cost Priority Queue After Expansion

1	S	0	A(1), G(12)
2	A	1	C(2), B(4), G(12)
3	C	2	D(3), B(4), G(4), G(12)
4	D	3	B(4), G(4), G(6), G(12)
5	G	4	Goal Reached

Cost Calculation of All Possible Paths

Path 1

$S \rightarrow G$

Cost

$$= 12$$

Path 2

$S \rightarrow A \rightarrow B \rightarrow D \rightarrow G$

Cost

$$= 1 + 3 + 3 + 3$$

$$= 10$$

Path 3

$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$

Cost

$$= 1 + 1 + 1 + 3$$

$$= 6$$

Path 4

$S \rightarrow A \rightarrow C \rightarrow G$

Cost

$$= 1 + 1 + 2$$

$$= 4$$

Therefore,

Minimum Cost = 4

$S \rightarrow A \rightarrow C \rightarrow G$

with a minimum transportation cost of 4. Therefore, Uniform Cost Search successfully provides the most economical route for package delivery, making it highly suitable for real-world logistics and navigation applications.